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INTERESTS Measurement validity in model evaluation — whether the benchmarks and automatic judges used to rank models are measuring what they claim to. Concretely: counterfactual auditing of (multimodal) LLM judges, benchmark design grounded in large-scale real human behavior rather than model self-play, and social and strategic reasoning in LLM agents.

DIRECTION One result recurs across my projects: **surface competence ≠ underlying cognitive competence**. In geolocation, a correct answer is not a correct reasoning process; in social deduction, persuasive language is not strategic correctness; in multimodal judging, a verdict that looks right is not one that used the right evidence.

The question I want to work on: why can a language model be this fluent and this capable at the surface, yet lack robust reasoning, strategy, and depth — what is missing between generating language and thinking? At three levels:

1. **Measure the gap.** Evaluation grounded in real human behavior and cognitive process, built to separate surface success from the capacity it stands in for.
2. **Explain it.** Mechanistic accounts of what inside the model produces the failure.
3. **Close it.** Interventions that change the underlying process rather than the benchmark number.

Longer term: what computational process turns language generation into reasoning, conceptual restructuring, and discovery — a science of machine thought, rather than a model that writes more like someone who thinks.

PromptChecker (*early stage*) — instructions arrive one at a time and rarely specify the problem on their own; a model that answers immediately is answering a question that was never fully asked. Reconstruct what the user actually needs before responding, and treat “not yet enough to act” as a first-class output.

EDUCATION **B.Eng. in Computer Science and Technology — Northeastern University, China**

2023 – 2027 (expected) · Shenyang, China

— CGPA **4.05 / 5.00** (90 / 100) · **Rank 5 / 100** in major cohort

— Research advising: Shiqi Zhao, Brain-Computer Interface and Hardware Acceleration Lab, since 2024

PUBLICATIONS * denotes equal contribution.

Beyond Survival: Evaluating LLMs in Social Deduction Games with Human-Aligned Strategies

Zirui Song*, Yuan Huang*, Junchang Liu*, et al.

EMNLP 2026 MAIN · co-first

author · arXiv:2510.11389 · Dataset

Geolocation with Real Human Gameplay Data: A Large-Scale Dataset and Human-Like Reasoning Framework

Zirui Song, Jingpu Yang, Yuan Huang, et al.

PREPRINT · arXiv:2502.13759 · Dataset · Code

EditJudgeBias: A Preservation-Gated Counterfactual Audit of MLLM Judges for Instruction-Based Image Editing

Yuan Huang — sole first author

IN PREPARATION · targeting ICLR 2027

RESEARCH

EditJudgeBias — counterfactual auditing of multimodal image-editing judges

2026 – present · Sole first author: conception, design, implementation, analysis, writing

Leaderboards for instruction-based image editing increasingly rank models with MLLM judges that are not themselves audited. This project makes “how much of the score is edit quality” measurable.

- **Method.** A single invariant runs through the pipeline: admit only perturbations that can be shown not to have changed edit quality. Preservation verification is an **admission gate** every injector and metric must pass — not a post-hoc check — with coverage reported layer by layer, and each cue carries a **zero-dose sham control** establishing a per-judge, per-metric floor so that null results are readable. (Preservation checking itself is not novel; making it a gate, and pairing it with sham floors, is the departure.)
- **Scale.** 13 cue conditions across four injection sites — protocol, pixel, content, prompt — plus a null control; 5 published judges plus an independent replication arm carrying its own BH family; base images from 5 public sources through a text / sha1 / ORB near-duplicate gate into 3 disjoint content pools. **97,973 judgments**, 1:1 with raw responses, 0.300% parse failure. 11 cues carry human-annotation legs, three rounds, $\kappa = 0.8930$ (read beside $\kappa_{\max} = 0.8930$).
- **Findings.** Protocol-level reordering alone — nothing else touched — flips decisions at **38.2× the judge's self-noise**. Five pixel/content cues depress across 5/5 judges; bandwagon is the only consistent inflator. Most effects are order-preserving depressions. **Rubric scoring buys absolute-score stability but not rank stability.** Of three mitigation layers only the protocol-level one works, and its mechanism is **abstention, not debiasing**; ensemble median is worse than nothing. A 5/5 null on a skin-tone attribute is readable only because same-batch positive controls fired — testing specificity alone would certify a never-alarming judge as best-calibrated (revocation sensitivity 0.032 vs 0.946 on identical data).
- **Verification.** 1,307 unit tests; 109 frozen artifacts each with sha256, producer, and an explicit “must not be read as” field; a single reconstruction path with byte-level comparison after full rebuild; a self-certifying delivery package (1,289 tests pass in-package, 112 rebuilt analysis tables byte-identical). The audit overturned four of my own conclusions, including a correction of one claim from 13/40 significant cells to 2/40 after adding clustering.

WereBench / WereAlign — human-aligned evaluation of LLM social reasoning

EMNLP 2026 Main · Co-first author; from project inception through submission

- Built a benchmark from **100+ hours** of professionally produced human social-deduction gameplay (**32.4M utterance tokens**, 30 roles, 15 rule variants, multimodal), using **winning-side MVP strategy trajectories as ground truth** rather than LLM self-play.
- Designed a 5-dimension verbal evaluation and 2 decision tasks; contributed to research design, experiments, evaluation, and iterative refinement.
- **Half of the evaluated models score below 0.50**; the best reaches **0.720**. Weakest dimensions are **deception reasoning** and **counterfactual trade-off** while persuasive generation stays strong — **fluency ≠ strategic correctness**.
- Two controlled interventions separated distinct failure paths, showing models can mistake another speaker's persuasive utterance for an instruction.

GeoComp / GeoCoT / GeoEval — geolocation from real human gameplay data

Preprint · Co-author (3 of 9)

- Contributed to a dataset from a public geo-guessing platform: **740K users, 2.7M locations, 25M human response records**, with human accuracy calibrating item difficulty.
- Multi-step geographic reasoning improved standard metrics by up to **25%**; scoring the **reasoning process** rather than only the final answer added a further **9%**.

HONORS

China National Scholarship — national level · 2023–2024

Northeastern University Comprehensive Scholarship · 2024–2026

Third Prize, National Finals, Lanqiao Cup Programming Competition · 2024

SKILLS

LANGUAGES

Python, C++, Java

ML / RESEARCH

PyTorch, Hugging Face Transformers

ENGLISH

IELTS Academic — taken 4 September 2026 (score to follow); CET-6 534

ALSO

Backend engineering intern, 2024 — Java / Spring Boot.